Yisen Wang

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Research Interests

Symbolic integration; Symbolic summation; Wilf–Zeilberger theory; Computer algebra.

Education

Sep. 2017 – Jun. 2021	**Beihang University** Bachelor's degree in Mathematics and Applied Mathematics
Sep. 2021 – Jun. 2026	**Chinese Academy of Sciences** Integrated M.Sc.–Ph.D. Degree in Applied Mathematics Supervisor: Shaoshi Chen
Oct. 2023 – May 2027	**Johannes Kepler University Linz** **RICAM, Austrian Academy of Sciences** Ph.D. Program in Computational Mathematics Supervisor: Christoph Koutschan

Publications **(Except for Publication 4, authors are listed in alphabetical order)**

1. Shaoshi Chen, Manuel Kauers, Christoph Koutschan, Xiuyun Li, Rong-Hua Wang and **Yisen Wang**. *Non-minimality of minimal telescopers explained by residues.* In Proceedings of the 2025 International Symposium on Symbolic and Algebraic Computation (ISSAC '25). ACM, New York, NY, USA, 70–78.
2. Shaoshi Chen, Christoph Koutschan and **Yisen Wang**. *How to generate all possible rational Wilf–Zeilberger forms?* Journal of Symbolic Computation, 132 (2026), Paper No. 102473, 23 pp.
3. Shaoshi Chen, Lixin Du, Hanqian Fang and **Yisen Wang**. *On the summability problem of multivariate rational functions in the mixed case.* In Proceedings of the 2026 International Symposium on Symbolic and Algebraic Computation (ISSAC '26). ACM, New York, NY, USA, 96–104.
4. Beibei Xiong, Hangyu Lv, Junqi Liu, **Yisen Wang**, Shaoshi Chen, Jianlin Wang, Zhengfeng Yang and Lihong Zhi. *Automated formal proofs of combinatorial identities via Wilf–Zeilberger guidance and LLMs.* In Proceedings of Forty-third International Conference on Machine Learning (ICML '26), Seoul, South Korea. PMLR 306, 2026.
5. Shaoshi Chen, David A. Cox and **Yisen Wang**. *Symbolic integration of differential forms: from Abel to Zeilberger.* 2026, To appear in Mathematics of Computation.

Talks

1. *Symbolic Computation in Abel's Addition Theorem for Algebraic Integrals.* CM 2023. Dalian, China.
2. *Generalized Hermite Reduction, Creative Telescoping and Definite Integration of D-Finite Functions.* Seminar Algebra and Discrete Mathematics. May 2024, JKU Linz, Austria.
3. *How to Generate All Possible Rational Wilf–Zeilberger Forms?* Aug. 2024, University of Vienna.
4. *WZ-Pairs in Combinatorial Identities.* Ph.D. Seminar. Sep. 2024, RICAM, Austrian Academy of Sciences.
5. *Non-Minimality of Minimal Telescopers Explained by Residues.* July, ISSAC 2025. Guanajuato, Mexico.
6. *Symbolic Integration of Differential Forms with Rational Coefficients.* Linz Algebra Research Day. Jun. 2026, RISC, Hagenberg im Mühlkreis, Austria.
7. *On the Summability Problem of Multivariate Rational Functions in the Mixed Case.* July, ISSAC 2026. Oldenburg, Germany.

Teaching Assistantships

Mar. 2025 – Jun. 2025	**University of Chinese Academy of Sciences** Teaching Assistance for Linear Algebra Course instructor: Ziming Li

Skills and Languages

Computer algebra system	Maple; Mathematica
Typesetting	LaTeX
Languages	Chinese (native); English (fluent)